Interactive Visualization for Statistical Modelling through a Shiny App in R

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Abstract—**The importance of analytics and visualization tools has been growing over the last decades to handle big data which steamed from all aspects of life. The focus of this paper was on visualization as a crucial tool in presenting complex raw data and modelling results to provide easy-to-understand actionable information that facilitate decision-making. However, limited research distinguished between "data visualization" and "model visualization", which has been clearly made in this paper. Furthermore, this paper aimed to shed light on the importance of interactive visualizations to compliment statistical data modelling using R and Shiny for its advanced capabilities. Specifically, a methodology has been proposed based on a hybrid development lifecycle that adopts the Agile Software Development Lifecycle and the Data Analytics Lifecycle. Finally, by presenting a case study to model the dynamics of COVID-19, it was found that R and Shiny alongside the proposed hybrid development lifecycle significantly reduced the amount of time required to build visually interactive applications. The reported results highlighted the effectiveness of the adopted approach in assisting and guiding researchers and developers in building interactive applications that leverage Big Data Analytics.**

Keywords—Data Analytics, Interactive Applications, R, Shiny, Visualization

I. Introduction

The overwhelming growth in the amount of data being generated has been a topic of interest for diverse explorations, research, and applications [1]. Putting the growth into perspective, the amount of data generated and consumed globally was 41 Zettabytes in 2019 and reached 64.2 Zettabytes in 2020, an increase of ~57% in just one year [2, 3]. Such large amounts of data, in terms of Volume, has been defined as Big Data (BD) [4], which soon after has been characterized by more traits such as Velocity and Variety, as introduced by Doug [5] to be later recognized as the 3V's of BD. Apart from the 3Vs, and the extended characteristics that other authors have explored in their pursuit to define BD, it is inarguable that BD has recently been growing to be indispensable to organizations [6].

As highlighted in the systematic literature review conducted by Sivarajah, Kamal, Irani, and Weerakkody [7] and demonstrated in various surveys and case studies [6, 8, 9, 10, 11], BD is exceptionally important to organizations as analysis conducted on such data, referred to as Big Data Analytics (BDA), can be beneficial in: (1) decision-making, (2) improving operational efficiency, (3) driving new revenue streams or even (4) gaining competitive advantages, and (5) improving resource allocation, to name a few. However, performing BDA is not an easy task, as there are various challenges that come with the high-volume, high-velocity, and high-variety of BD, among other aspects and challenges [7].

Among the challenges and aspects of using BD and performing BDA is visualization which has been growing in importance due to its ability to simplify findings and to enhance decision-making [12]. Despite the importance of visualization, to the best of the authors' knowledge, there was no relevant study that clearly differentiated between the various types of visualizations with an exception to the interesting perspectives and demonstrations presented by Wickham, Cook and Hofmann [13] who provided only a lose definition to the terms data visualization and model visualization. Accordingly, this paper aimed to fill this gap by providing a clear distinction between data visualization (data-vis) and model visualization (model-vis).

In general, visualization revolves around visually presenting key information or knowledge in an instinctive and effective manner [14]. With a focus on data-vis, which can be performed directly on raw or processed data, or as part of the BDA process, uses graphical representations to illustrate patterns, trends, and relationships among elements of the data [15], essentially the visualizations are extracted from and/or about the data themselves. Examples of data-vis include using line graphs to understand the relationship between two or more variables or scatter plots to identify simple high-level clusters (groups) within the dataset [15]. Therefore, data-vis is crucial as it facilitates (1) easier understanding of the data themselves, (2) simple exploratory analysis to identify trends and patterns within the data, and (3) can assist in more efficient decision-making [16]. In an executive report published by International Business Machines (IBM) [17], the advanced capability of data-vis was among the top three vitally required capabilities when dealing with BD, in both the software tools and human resource skills – primarily due its ability to transform the high-variety of BD into easy-to-understand, actionable information [18].

On the other hand, model-vis focuses on presenting the results of statistical models [19] or other complex models including Machine Learning (ML) black box models [20]. Model-vis can also be referred to as presenting the data in the model space, which is commonly practiced – for example, presenting predicted versus actual data points [13]. Model-vis is beneficial as it visually aids in the understanding of predictions by assessing model assumptions and goodness-of- fit [19], interpretation and communication of results [21], decision-making [12], or even assist in the development of new theories – which without visualization would be difficult and high-effort demanding to interpret a lot of data and numbers, or more complex outputs generated by the statistical models or algorithms [16]. The need to clearly distinguish between data-vis and model-vis is essential and demonstrated by authors in [13] as they clearly described the conversion of data-vis methods to model-vis methods as "not always straightforward", deducing that the two terms differ.

While the benefits and importance of the two types of visualizations have been explored heavily by researchers, it is also important to note the difference between static and interactive visualizations. Static visualizations graphically present findings that do not change over time [22], hence only depict the situation or viewpoint at the time they were generated. On the contrary, interactive visualizations allow the user to explore, interact and directly control the graphical representations generated; usually done by allowing the user to modify parameters that are mapped to underlying configurations applied on the data or models used for the visualizations [23]. Interactive visualizations are beneficial in various ways such as allowing for more than one viewpoint to be generated, unlike the single viewpoint in static visualizations, and effective and efficient identification of relationships and trends as the user can change parameters and instantly observe their impacts. In addition, interactive visualizations can simplify and present complex data, and assist in dynamic data storytelling, which is facilitated by changing parameters and exploring multiple viewpoints [24]. Nonetheless, static visualizations are also important and useful when dealing with less complex data stories, or when the requirement is to focus on portraying a predetermined view rather than exploration [23]. Understanding the difference between the two approaches is important, as the selection of either approach should be based on the purpose of the visualization to avoid over- or under-engineering the solution and failing to achieve the expected outcomes [25].

Based on the above discussion, this paper aimed to demonstrate the benefits that can be reaped by leveraging interactive model-vis using Shiny [26], a publicly available package that allows users to build interactive web application directly out of R, a programming language and environment for statistical computing [27]. Though there are several tools that allow interactive visualization such as Tableau [28] or Microsoft's Power BI [29], the combination of Shiny and R has been selected for R's powerful statistical backend which includes ready-made packages with various statistical algorithms and the ability to build extremely customized business logic (i.e., code) and leverage Shiny's extensive and customizable frontend capabilities to present a customized interactive web applications tailored for the business's requirements. As mentioned by authors in [12], there is no single tool that can be determined as the best, as the selection should be based on the requirements of the solution. However, unlike other papers which focused on comparing visualization tools, this paper aimed to demonstrate the benefits of using Shiny and R (R/Shiny), as similarly conducted by Wojciechowski, Hopkins, and Upton [30].

There are various fields and applications which authors have explored using R/Shiny. For example, authors in [30] demonstrated using R/Shiny in the field of pharmacometrics to build various interactive applications and concluded that elegant, flexible and informative interfaces and model simulations can be quickly produced. Furthermore, authors in [31] developed an advanced interactive R/Shiny application, called shinyCircos, to simplify the process of creating Circos plots, addressing the limitations found in other tools that required coding knowledge/skills or were difficult to install and use. Additionally, authors in [32] used R/Shiny to create a new innovative tool, called robvis, with the explicit purpose of visualizing risk-of-bias results – in which the authors attributed their success to the availability of unique and powerful features provided by the combination of R/Shiny.

Moreover, in recent years, specifically since 2019 with the emergence of the global coronavirus disease (COVID-19) pandemic, several authors and developers have turned to BDA, statistical modelling, and visualizations, and building interactive web applications to assist hospitals and governments to respond to the global outbreak. Weissman et al. [33] developed an interactive web application that allows the users to perform an infinite number of simulations by adjusting various model parameters and reflecting instant updates on visualizations – allowing the user to predict hospital capacity needs during the COVID-19 pandemic and efficiently take time-sensitive critical decisions. Similarly, the state of California, in the United States of America, uses R/Shiny to visualize the current and short-term forecasted spread of COVID-19, as well as generating multiple scenarios for the spread – and by using R/Shiny, the interactive application was brought to life in a matter of weeks, which is considered exceptionally fast in comparison to the normal/traditional timeline required for application development and deployment [34].

With COVID-19 being a critical global pandemic causing severe repercussions on the global economy [35] in addition to the lives lost, it is critical to use BD, and tools and techniques (i.e., R/Shiny, BDA, and statistical models) to help and accelerate the containment of the spread of the disease and reduce the damage caused by the pandemic [36]. To this end, this paper demonstrated the advantages of building interactive R/Shiny applications by presenting a case study for the dynamics of COVID-19 cases over time. Though the accuracy and validity of the statistical models are highly important to ensure that correct decisions have been taken, it is not within the scope of this paper and hence basic statistical models have been used to simply demonstrate the ability of building interactive applications based on ready-made packages within R, and to show the advantages of interactively visualizing the model outcomes.

II. METHODOLOGY

In this paper, an interactive web application has been developed using R/Shiny which could be used with statistical models to facilitate answering the following questions:

- How well does the model capture the dynamics of the provided data?
- What is the effect of changing model parameters on the model outcomes?

The interactive web application has been built using a hybrid development lifecycle, which constitutes the proposed methodology, to serve the purpose of this paper and to ensure performing data analytics by leveraging the advantages of R/Shiny. The hybrid development lifecycle has been formulated by combining the Agile Software Development Lifecycle and the Data Analytics Lifecycle which are detailed in what follows.

A. Agile Software Development Lifecycle

The Agile Software Development Lifecycle (ASDLC) was designed to allow organizations to quickly build and deploy applications (i.e., software), and hence, leveraging collaborations in all development phases, decision making and prioritization of requirements and development iterations. The ASDLC is based on an iterative approach, in which each iteration (often referred to as "sprint" or "cycle") focuses on a limited set of features/requirements and carrying out the full software development lifecycle in shorter and condensed periods of time including the six phases demonstrated by the upper panel of Fig. 1 represented by Requirements Collections, Design and Prototyping, Development, Quality Assurance and Review to User Acceptance Testing (UAT), and finally Release. By focusing on smaller sets of requirements, organization and teams can develop systems in weeks, and continue to enhance and deploy new updates with ever sprint of the Agile SDLC [37, 38, 39].

B. *Data Analytics Lifecycle*

The Data Analytics Lifecycle (DALC) is rather a scientific methodology devised to guide the process of data analysis. The DALC follows an iterative approach between each phase, allowing the movement back and forth between steps of the process, which facilitates "trial & error" and improvements to eventually reach a good end result. The process usually consists of six main phases as demonstrated by the lower panel of Fig. 1 and represented by Discovery – in which the researcher examines what data are available and what are the requirements, Data Preparation – the data are prepared to meet the requirements, Model Planning – identifying the models, statistical techniques, or algorithms to be used, Model Building – the development and execution of the planned analytical/statistical models, Communication of Results – to verify the accuracy and validity of the models, and finally Operationalizing – to deploy the model to be used by the organization and fulfil the requirements. The DALC is important as it facilitates and ensures the correct models, with validated results are developed and deployed to users to achieve and fulfil the requirements [40, 41, 42].

C. *Hybrid Development Lifecycle*

To meet the main purpose of this paper in developing a visually interactive application that can perform data analysis, a hybrid development lifecycle was proposed which consists of five phases as a result of merging the phases of both ASDLC and DALC. Therefore, the first phase is meant to perform data discovery as well as requirements collection to define what is needed from the interactive application. Following this, the design and prototype phase which includes steps of data preparation, model planning, designing and prototyping the user interface based on the requirements and planned models. The third phase conveys development towards building the actual data analytical models and application in code. Then, communication of results which represents the fourth phase that includes the quality assurance and UAT of the application to ensure the model results are correct and as per the requirements. Finally, based on the communicated results and UAT, the final phase of release is to be carried out to deploy and operationalize the interactive application with its data analytics and interactive visualizations. Fig. 1 illustrates how the two lifecycles (ASDLC and DALC) were overlaid, mapped, and combined into the final proposed hybrid development lifecycle as depicted by Fig. 2.

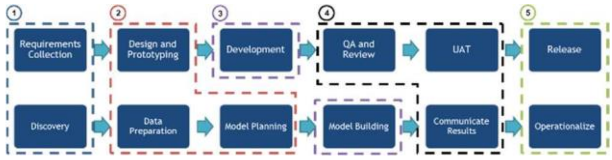


Fig.1. Mapping of ASDLC to DALC

Fig.2. Proposed and Adopted Hybrid Development Lifecycle

III. Results and Discussion

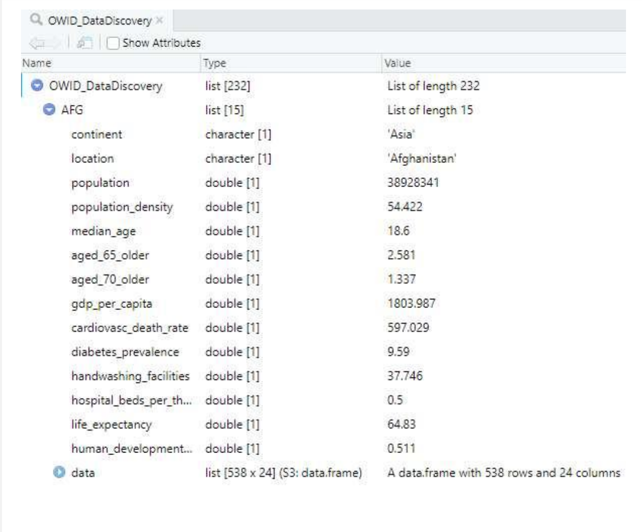
This section presents a case study to depict the dynamics of COVID-19 new cases using simple statistical models. Furthermore, each stage of the proposed hybrid development lifecycle is detailed.

A. *Requirements Collection*

This step includes identifying the requirements of the application and data discovery. For the case study under consideration, the requirement is fitting a model to capture the dynamics of "new COVID-19 cases" using smoothing splines and visualize the fitted model.

It is worth mentioning that smoothing splines represent a family of non-linear non-parametric statistical regression models which have been used in this work for their flexibility to identify interesting trends. Furthermore, using interactive visualizations will enhance the understanding of the trends captured by the adopted statistical model, and facilitate judgment on the model fit using different values of model parameters.

The dataset was collected from the publicly available dataset provided by Our World in Data [43], obtained as JavaScript Object Notation (JSON) format. The dataset contains 38 attributes (including the date and new cases for that date) for each country, from the first recorded COVID-19 case to the time of writing this paper – resulting in a dataset with over 100,000 records. As part of data discovery, the dataset was identified to fulfil the case study requirements, as illustrated in Fig. 3.



OWID_DataDiscovery ×

Show Attributes

Name	Type	Value
OWID_DataDiscovery	list [232]	List of length 232
AFG	list [15]	List of length 15
continent	character [1]	'Asia'
location	character [1]	'Afghanistan'
population	double [1]	38928341
population_density	double [1]	54.422
median_age	double [1]	18.6
aged_65_older	double [1]	2.581
aged_70_older	double [1]	1.337
gdp_per_capita	double [1]	1803.987
cardiovasc_death_rate	double [1]	597.029
diabetes_prevalence	double [1]	9.59
handwashing_facilities	double [1]	37.746
hospital_beds_per_th...	double [1]	0.5
life_expectancy	double [1]	64.83
human_development...	double [1]	0.511
data	list [538 x 24] (S3: data.frame)	A data.frame with 538 rows and 24 columns

Fig.3. Sample of the raw dataset

The interactive application was developed to allow the user to select the country and to use a slider to adjust the model parameter of training-testing split to instantly reflect the updated model visualization.

B. *Design and Prototyping*

This step includes designing and prototyping the application as well as data preparation and model planning. Specifically, Fig. 4 presents the outline of the planned application in terms of User Interface (UI) using the Shiny package, while system functionality is shown by Fig. 5 which presents the planned flow of the application to assist in the development of the needed functions which were identified as part of model planning, ensuring that correct steps were taken to build the needed model.

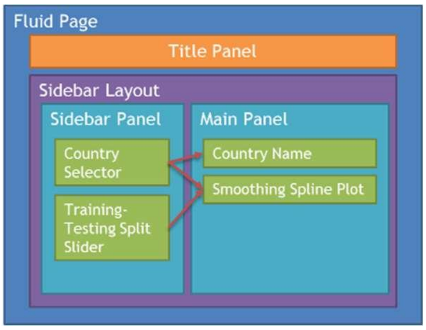


Fig.4. Planned Interactive Application User Interface Design

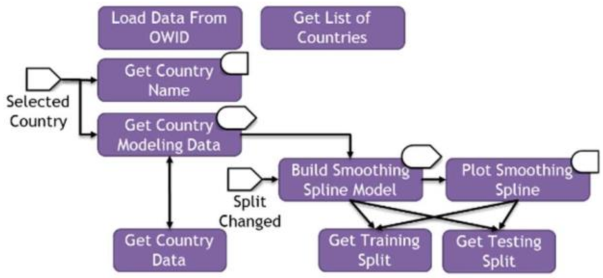


Fig.5. Planned Application Flow and Functions

In terms of data preparation, then the dataset needs to be prepared and cleaned by extracting only the essential information for modelling, which are the date and number of cases for that date. The “date” was transformed to a day counter to resemble time.

C. *Development*

This step includes the actual development of the interactive application and building the analytical/statistical model. Since R/Shiny was used, three main files were created. Specifically, the first file contains the user interface, the second file contains the server logic, while the third file contains all the functions planned and designed in Fig. 5. Working through these files has the advantage of simplify the code. Indeed, the development of these three files, and eventually the visually interactive application took less than four hours to build and finalize which is an exceptional time in comparison with traditional web applications, making the proposal a useful tool for building an interactive application to reinforce model visualization.

D. *Communication of Results*

This step includes testing the interactive application and performing quality assurance and UAT to ensure no errors or bugs occur in the application. Once confirmed, the results of the model are to be communicated with subject matter experts for verification and validation of the model’s accuracy.

However, given the novelty of the COVID-19 disease and the highly variable future of the pandemic, it is difficult to confirm the validity of predictions until it occurs. Hence, this paper depended on visualizing the trend against the current data as per the training-testing split defined by the user, with the final application presented in Fig. 6, which presents three different models generated on-spot which illustrates the usefulness and efficiency of interactive visualization.

It is apparent from the plots in Fig. 6 that the model fits data well, but poor predictions have been generated with the adopted model, which can be further improved by using different models, for example a mixture of models, to capture the significant spike (specific to the selected country, Bahrain) in cases near the end of the timeframe of the case study. It is also interesting to note careful selection of parameters can determine the goodness-of-fit depending on the nature of the dataset used, for example, using fixed parameters against using time-varying parameters.

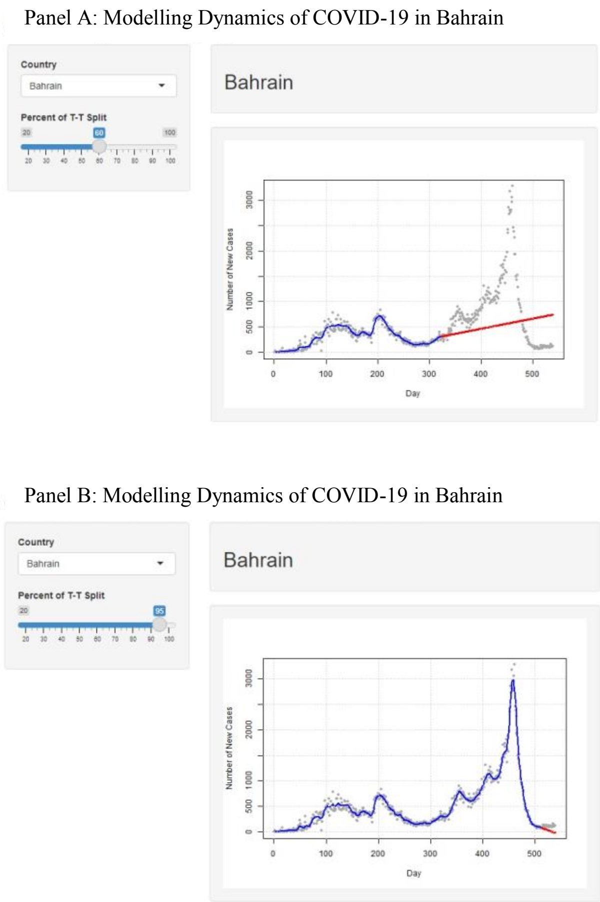


Fig. 6. Screenshots of the built application using R/Shiny

E. Release (Operationalize)

This step includes deploying the application to a production environment as well as operationalizing and relying on the results of the statistical model in real-world applications. However, this step was considered being out of the scope of this paper.

IV. Conclusion

In this paper, a clear distinction has been made between data visualization and model visualization, with both types being crucial for Big Data Analytics.

Moreover, an application has been developed for interactive visualizations which represents a beneficial analytical tool when the desire of the research is data explorations or to understand the impact of parameters on modelling outcomes. This target has been met by using the advanced features available in the programming language R and the Shiny package, beside the proposed methodology which has been devised based on hybrid development adopting the Agile Software Development Lifecycle and the Data Analytics Lifecycle.

The proposed methodology has been illustrated in a case study to capture the dynamics of COVID-19 using smoothing splines, to demonstrate the capabilities of efficiently and effectively building statistical models with interactive model visualization using R/Shiny. The outcomes of the proposed methodology resulted in a finalized application, built in a short time (less than four hours). Moreover, the interactive visualizations created showed how the model fit adjusts to the parameters instantly with minimal development effort required and maximum outcome achieved which could assist decision makers and generally demonstrated the efficiency and effectiveness of R/Shiny.

However, it is worth mentioning that this paper did not validate the accuracy of the statistical models in capturing the dynamics of COVID-19 as there were sophisticated models presented in the relevant literature for this purpose and interested readers can refer to them [33, 44, 45, 46, 47, 48]. The primary aim of this paper was to shed light on the efficiency and effectiveness of using R/Shiny to build interactive model-visualizations, and hence, the implemented statistical models (smoothing splines) were just used for illustration purposes.

Moreover, the proposed methodology is based on the Agile Software Development Lifecycle, which by nature is generic and serves as principles rather than steps “written in stone”, hence a proposed future study can be carried out to compare efficiency and effectiveness of a hybrid of the Data Analytics Lifecycle with Scrum Software Development Lifecycle against the Kanban Software Development Lifecycle, which are slightly different approaches of the Agile.

As a future work, a systematic comparison can be conducted between R/Shiny against other development techniques (for example Python/Flask or Tableau). Furthermore, in terms of the proposed methodology, further research and case studies can be carried out to verify and validate its effectiveness across different domains and applications.

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